# An Evaluation of The Performance of Lightweight Language Models on NL2SQL tasks

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## 1 ABSTRACT

Since their introduction as a consumer product in 2022, transformer-based large language models have been used for automating many kinds of digital tasks. This includes natural language to code and SQL tasks, which have exploded in research and development in the last few years. While a large proportion of LLM use is likely done using proprietary, closed source, and large footprint foundation models hosted by SAAS providers like Anthropic or OpenAI, an increasing amount of open-source models are also entering the market. These models have demonstrated promise on several different kinds of tasks, such as code generation using Python [1]. In this paper, we aim to provide an overview of how several of the smaller open source models perform when faced with elementary NL2SQL tasks on the Spider-Dev question dataset.

## 2 INTRODUCTION

In recent years, there has been a significant amount of high performing open source LLMs available from both research institutions and tech companies [3]. While many of the top-ranking models require expensive and complex hardware setups to run as their parameter counts run in the hundreds of billions or even trillions, many are either small footprint by design or distilled versions of larger source models.

In these cases, consumer level hardware should be able to run the models using tools like localLLama, enabling their use offline and without dependency on external SAAS providers (for purposes of this paper, we define consumer level hardware to be current generation Nvidia RTX chips or other equivalents, able to run models in the 20-30b parameter range or smaller at 8 or 16b quantization).

At the same time, there has been significant development in the field of NL2SQL tools. While previous generations of NL2SQL tools depended on hardcoded rules or deep neural networks [2], the emergence of LLMs has enabled a whole new paradigm of NL2SQL tools based on them. In fact, in their MVP state, a LLM itself can serve as a NL2SQL tool on its own.


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## 3 TESTING METHODOLOGY

## 4 RESULTS

## 5 ANALYSIS

## 6 CONCLUSION

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